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Electrify and Amplify: How Bits and Bytes Drive Demand for Watts and Volts

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Before I joined 22V, I read a research report by their commodities expert, Colin Fenton, titled *What Glut?* Colin and I have known each other for years, and I've always respected his independent thinking—but this report stood out. I had been consuming a large number of podcasts following the “Drill Baby Drill” election outcome and felt this was the perfect setup for a long AI theme that ran counter to consensus thinking.

Colin's research challenged the widespread belief in a 2025 oil supply glut by pointing to how OPEC+'s revised production plan—and likely forecast revisions from the EIA and IEA—could tighten the global oil balance significantly. He argued that the IEA's surplus projection was likely overstated due to underestimated demand and overly optimistic supply assumptions, especially when measured against historical norms, low starting inventories, and more balanced forecasts from the EIA and 22V. Ultimately, the report emphasized that the glut narrative was built on fragile assumptions and could be quickly reversed—setting up a potential bullish surprise for oil markets in 2025.

In January, he followed up that with a video titled, *Abandoned Alpha*. In it he said, “We contend that, over the next two years, all investors will better come to understand oil and gas are not “stranded assets” and it is a serious strategic mistake to divest from them in portfolios. In truth, oil and gas firms offer “abandoned alpha” to embrace. With both fuel and non-fuel applications, hydrocarbons are unavoidable markets. Their responsible development and use—with just rewards for their creditors and investors—will be essential for the sustained well-being and improvement of all sectors and all economies for as long as human civilization endures.”

At the same time Colin published his piece, I had been producing content on higher future energy prices—but from a completely different angle. My focus was on the insatiable demand for power driven by the rise of artificial intelligence.

While much of the attention in D.C. right now is focused on tariffs, let's go back to before the announcement in the Rose Garden on April 2nd. The best performing sector in the SPX, up 10% for the year, was the energy sector with WTI trading at 72 and Dec 25 nat gas at 5.15. A week later on April 9th, the lows were 55 and 4.40 as markets built in the end of the world.

Also on April 9th, the House of Representatives held a hearing on the future of AI technology and American competitiveness. The goal was to assess how the United States can lead in the global AI race while protecting national security, economic competitiveness, and democratic values. Lawmakers and industry leaders discussed the policies, investments, and regulatory frameworks needed to responsibly scale AI across sectors such as healthcare, defense, and manufacturing. At the core of this conversation was one undeniable truth: success hinges on the U.S. meeting the enormous and fast-growing energy demands required to power AI infrastructure—especially data centers.

When investors think about AI, they often focus on its outputs—the incredible breakthroughs it promises. From humanoid robots and self-driving cars to longevity treatments and disease cures, AI is usually framed around its potential to transform lives. But what's often overlooked—and was a central theme in the hearing—is the staggering amount of energy needed to make that future real. Energy isn't a side issue; it's the backbone of the AI revolution.

Eric Schmidt, former CEO of Google, drove this point home at the hearing: “People are planning 10 gigawatt data centers now... an average nuclear power plant in the United States is one gigawatt.” He warned that AI-driven data centers could

require 90 gigawatts of new power by 2030—equivalent to building 90 nuclear plants—as AI moves beyond chatbots to full-scale reasoning and planning systems.

This surge in demand places tremendous pressure on America’s existing power grid, prompting urgent calls for reform. “We are behind,” said Manish Bhatia, Executive VP at Micron Technology, noting that even “fractions of a second of power drop” can cost semiconductor fabs tens or hundreds of millions of dollars. He shared that Micron’s U.S. operations alone are expected to require 2 gigawatts of energy by 2040—and warned that permitting delays and grid constraints could derail essential manufacturing projects.

Throughout the hearing, multiple witnesses advocated for an “all of the above” energy strategy. Schmidt underscored the urgency: “What we need from you... is energy in all forms—renewable, non-renewable—whatever it is, it needs to be there, and it needs to be quickly.” David Turk, former Deputy Secretary of Energy, agreed, stressing that while renewables are the fastest and cheapest to deploy, advanced nuclear, geothermal, and all other sources must be part of the solution. “Housing as many AI data centers as possible,” he said, “is both an economic and a national security imperative.”

The panelists warned that red tape, supply chain delays, and policy uncertainty—especially surrounding tariffs and clean energy incentives—could significantly undercut AI progress. “The delays are crazy,” Schmidt said, referring to 18-year permitting timelines for new transmission lines. Bhatia added that redundant permitting at both the federal and state levels creates costly slowdowns. The consensus was clear: if the U.S. wants to maintain its lead in the global AI race, coordinated and immediate investments in energy generation and infrastructure are non-negotiable.

As investors begin again to focus on durable themes post Liberation Day, this has made this topic more important. Regardless of where the tariffs settle in percentages, the world has changed. Globalization has reshaped where labor, manufacturing and revenues come from to impact investments. Going forward, you want to find durable themes that are subsidized to some degree by the governments and in particular have competition for it. AI and power and that relationship has no borders. Every country needs both of them for the future and they all know it. As nationalism rises post tariffs and countries focus on competition, these themes will remain structural demand growth stories. In this deep dive paper, I’ll explore the historical relationship between innovation and power—and why I believe that over the next decade, the balance between the two will shift dramatically. As AI becomes increasingly central to geopolitics and economic growth, energy will emerge as a key strategic asset. The unwinding of positions across markets has created a rare opportunity for investors to rethink how they approach the energy sector—and to identify power-related opportunities with growth potential that could rival what we’ve seen with companies like Nvidia. The weighting differential in the SPX and the World indices between technology and power/energy does not currently reflect the potential energy needs to fuel AI in the future which creates the opportunity today.

Executive Summary

- **AI’s Energy Surge:** The rapid proliferation of artificial intelligence is driving an unprecedented surge in electricity demand. Data centers worldwide are on track to **double their power consumption by 2030 to ~945 TWh** – about equal to Japan’s entire current electrical use . In advanced economies like the U.S., **AI-related computing could account for nearly half of new power demand this decade**. This represents a **seismic shift** in energy usage patterns reminiscent of past industrial revolutions.
- **Historical Parallels:** Major technological leaps have always triggered spikes in energy consumption. The Industrial Revolution saw **per-capita energy use triple** in 18th–19th century Britain , fueled by coal and mechanization. Similarly, the digital revolution of the late 20th century vastly expanded electricity use as economies electrified. These precedents underscore that **innovation runs on energy (“power as electrons”)**, with electricity becoming the indispensable medium for progress.
- **From Data to Experience:** AI’s energy footprint is expanding from cloud data centers to real-world machines. **Training large AI models is power-intensive**, often requiring megawatt-hours of electricity, while **real-time AI inference** in products like robots and autonomous vehicles is multiplying energy needs at the edge. A single **self-driving car’s AI system can draw 0.5–2.5 kW** of continuous power , and fleets of EVs and robots learning from experience will add significant load to grids.

- **Rising Demand Outpacing Forecasts:** Consensus forecasts anticipate strong baseline growth in global and U.S. energy demand, but **AI is poised to amplify these beyond official projections**. The IEA projects global electricity use will grow ~4% annually through 2027 (adding more than Japan’s consumption each year) , *before* accounting for AI’s full impact. U.S. power demand, after a decade of stagnation, could **rise 15–20% this decade** largely due to data centers and electrification . Without adjustments, **demand growth may outrun current supply plans**, creating a potential gap.
- **Policy Constraints and Supply Risks:** At the same time, climate policies and market signals have constrained investment in traditional fossil fuel energy sources. Global upstream oil & gas spending remains below pre-2015 levels, raising warnings that **underinvestment could lead to supply shortfalls and price volatility** if demand stays strong . Many governments push for decarbonization and have canceled or curtailed coal, oil, and gas projects. This *mismatch* – surging demand vs. limited new supply – **risks energy security imbalances over the next 5–10 years**.
- **Winners in the Energy Complex:** In the near term, **natural gas and renewables are poised to benefit** as critical sources of new electricity for AI. Gas-fired power provides the on-demand capacity needed for data centers , and renewables (solar, wind) will capture investment due to cost-competitiveness and climate mandates. **Nuclear power** may experience a renaissance as a stable, low-carbon supply for AI-driven demand. Likewise, **energy storage solutions** (batteries, etc.) and grid infrastructure upgrades stand to gain increased urgency and funding to manage higher loads and intermittent supply.
- **Investment Implications:** AI’s energy revolution creates both opportunities and risks across markets. **Opportunities:** Utilities and independent power producers positioned to add capacity (especially in gas, renewables, and potentially nuclear) could see revenue growth. Companies supplying the AI infrastructure value chain – from **chipmakers focused on energy-efficient AI processors** to data center REITs, electrical equipment manufacturers, and battery/storage producers – are positioned to outperform as capital flows into expanding capacity. **Risks:** Energy-intensive industries could face higher input costs, and economy-wide inflationary pressures may rise if electricity prices spike. Policy and regulatory interventions (such as data center efficiency standards or carbon costs) are another wildcard for investors. Overall, **the nexus of AI and energy will reshape capital allocation**, favoring sectors that enable reliable power delivery and efficiency for the AI age.
- **Strategic Outlook:** Businesses and investors should prepare for a new phase of the energy transition where **AI-related demand becomes a major driver of market dynamics**. Strategic investment in power generation, grid resiliency, and AI-optimized infrastructure will be essential to support this growth. Those who anticipate the coming “**AI power crunch**” – and invest in mitigating it – stand to benefit, while those caught off guard may face disruptions. The following report provides an in-depth analysis of these trends, historical context, sector impacts, and actionable insights for investors.

Introduction and Historical Overview

The rise of artificial intelligence is not only a technological phenomenon but also an energy story. As AI algorithms proliferate and increasingly integrate into everything from cloud computing to autonomous machines, they are fundamentally altering the demand profile for energy – especially electricity. This report examines the economic and market implications of AI’s rapid growth on global energy demand and infrastructure, through the lens of history, current trends, and future forecasts. We begin by looking backward: how have past technology revolutions impacted energy consumption, and what lessons do they offer for today’s AI-driven boom?

Technological Revolutions Fueled by Energy: History shows that transformative innovations go hand-in-hand with surges in energy use. During the **First Industrial Revolution (late 18th to 19th century)**, Britain and other early-industrializing nations saw explosive growth in energy consumption. In England and Wales, total energy supply jumped **30-fold from 1560 to 1860**, driven by a shift from wood to coal and a six-fold population rise . Even on a per-person basis, “**useful energy**” **consumption roughly tripled** from about 1 MWh per capita in the 1600s to over 3 MWh by the mid-1800s . Steam engines and mechanized factories vastly increased the demand for coal power, marking an *unprecedented increase* in humanity’s energy appetite.

Later, the **electrification and oil age** of the 20th century – sometimes called the Second Industrial Revolution – again accelerated energy demand. Widespread electrification enabled mass production, lighting, and appliances, causing electricity to become the backbone of modern economies. For example, in the United States, total electricity generation rose from virtually zero in 1900 to over 4,000 TWh by 2000, a >4,000-fold increase, reflecting how essential electric power became to industry and daily life. Each wave of innovation (railroads, telegraph, automobiles, computing, etc.) was underpinned by greater energy consumption, whether in the form of coal, oil, or electricity.

“Power as Electrons” – Electricity’s Central Role: A key theme emerging from historical trends is the centrality of **electricity as the medium of innovation**. While the first industrial era was about converting heat (from burning coal) into mechanical work, the current era is about converting electrons into intelligent work. Advances in digital computing, telecommunications, and now AI all fundamentally rely on electrical energy. This concept of *powering progress with electrons* means that as technologies become more digital and data-centric, their energy source is almost exclusively electricity. For instance, the Information Age has seen the rise of data centers, semiconductor fabs, and millions of personal devices – all drawing on the electric grid. Global electricity consumption roughly **doubled from 2000 to 2020** amid the expansion of the internet, mobile devices, and cloud computing. In short, **technological progress has been inexorably linked to rising electrical energy use**.

Parallels to Today’s AI Boom: The current explosion of AI capabilities – from machine learning algorithms to intelligent robots – appears to be following this historical pattern. We are entering a phase where **AI development = increased energy demand**, especially electricity. Just as the advent of steam power or widespread computing created new energy needs, AI’s growth is beginning to stress existing energy infrastructure. The difference now is the *speed* and *scale* at which this is happening: AI adoption is accelerating on a global scale within years, whereas past energy demand booms unfolded over decades. Analysts are increasingly viewing AI through the lens of an industrial revolution in its own right – sometimes dubbed a “Fourth Industrial Revolution” – one that could trigger a significant uptick in power requirements.

In the sections that follow, we will delve into how AI’s energy needs are manifesting today, how they are expected to shape future demand globally and in the U.S., what constraints current energy policy imposes, which energy sectors stand to gain or struggle, and what this all means for investors and businesses. First, we clarify the nature of AI’s energy consumption – distinguishing the demands of massive cloud AI computations from the emerging **real-world, experiential AI** in robotics and autonomous systems.

The AI Shift: From Data to Experience

AI workloads are evolving, and so is their energy profile. Traditionally, AI development has been **data-driven and cloud-centric** – training algorithms on large datasets in energy-hungry data centers. Now, we are witnessing a shift toward **real-world experiential AI**, where intelligence is embedded in physical devices (robots, autonomous vehicles, drones, smart appliances) that learn from interaction with the environment. This transition from purely digital AI (in servers) to AI in the physical world carries significant implications for power and energy requirements.

Powering AI Training vs. Inference: It’s important to distinguish between two phases of AI computation – **training and inference** – as they have different energy footprints. *Training* an advanced AI model (for example, a large neural network like GPT-4 or a deep reinforcement learning agent) is extremely computationally intensive. It involves running billions of mathematical operations on specialized hardware (GPUs or TPUs) over days or weeks. This consumes **vast amounts of electricity**: recent estimates suggest training a single state-of-the-art model can use **hundreds of megawatt-hours (MWh)** of energy, roughly equivalent to the annual electricity use of dozens of U.S. homes. For instance, one analysis by Goldman Sachs noted that **a single ChatGPT query consumes about 2.9 Wh** (watt-hours) of energy, nearly **10× the energy for a typical Google search (0.3 Wh)**. Extrapolated over millions of queries, plus the training runs behind such models, it’s clear AI computation is far from trivial in power terms.

Once an AI model is trained, *inference* refers to using that model to make predictions or decisions (e.g. an AI answering questions or recognizing images). Inference per operation is usually less compute-intensive than training, but it happens far more frequently – potentially millions or billions of times across user applications. The collective energy for inference can therefore also be enormous, especially as AI is deployed at scale. Notably, **breakthroughs in AI efficiency don’t necessarily reduce energy demand – they often enable larger models and broader use cases**. As McKinsey observes, when computing power becomes cheaper or more efficient, AI researchers tend to utilize those gains to run

even more complex models rather than saving energy. In essence, **AI's appetite for compute (and thus power) keeps growing** in line with its capabilities.

To emphasize this point, during the April 9, 2025 House hearing, **Dr. Eric Schmidt** emphasized that the energy demands of AI are escalating rapidly—not just because of chatbot-like interactions, but because of the emergence of **reasoning and inference-based AI systems**, which require vastly more computational power.

Here's what he said:

"You think of AI as ChatGPT, but what it really is, is a reasoning and planning system that we've never seen before."

He explained that these next-generation models, which go beyond pattern recognition and into decision-making and planning, are **exponentially more energy-intensive**. Schmidt stressed that the compute and power requirements for these models are industrial in scale:

"What's happening at the moment in our industry is that we're very quickly developing AI programmers and AI mathematicians. These algorithms will need a lot more computation than we've ever had—they're going to need a lot more energy."

He underscored that training and running these models—especially for inference at scale—will drive unprecedented electricity demand, estimating that **data centers could eventually consume the majority of U.S. power**:

"Many people think the demand for energy from our industry will go from 3% to 99% of total generation. One of the most likely estimates is that data centers will require an additional 29 gigawatts of power by 2027 and 67 gigawatts by 2030."

Schmidt concluded that to meet this challenge, the U.S. must deploy **"energy in all forms—renewable, non-renewable, whatever it needs to be there—and it needs to be quickly."**

An important recent development in AI efficiency—and its paradoxical impact on energy demand—is the release of **DeepSeek R2** in January. Investors initially greeted this new model with panic that has helped drive the AI unwind this year. DeepSeek demonstrated significant gains in compute efficiency compared to other large-scale AI systems. By optimizing parameter usage and leveraging advanced compression techniques, DeepSeek R2 successfully cut power consumption for specialized tasks and promised to reduce the cost of running complex AI workloads. This led some market observers to predict a plateau in the overall energy demand from AI, assuming that DeepSeek-like innovations would propagate industry-wide.

However, the open-source nature and low entry barriers of DeepSeek R2 quickly flipped that narrative. While individual workloads became more efficient, the model's ease of deployment and customization dramatically **broadened** AI adoption—spurring a wave of new projects, startups, and hobbyists who could suddenly afford advanced AI reasoning at scale. Notably, several rival AI LLM firms—ranging from boutique labs to well-known platforms like **MosaicML**, **EleutherAI**, and various emerging open-source collectives on **Hugging Face**—responded by *accelerating* the release of their own "reasoning-optimized" models. In some cases, they explicitly leveraged DeepSeek R2's compression libraries and inference optimizations to ship competing LLMs months ahead of prior roadmaps. By the end of Q1, the market saw a surge of newly available open-source reasoning engines, each promising superior throughput, step-by-step logic, or domain-specific intelligence.

This flood of offerings paradoxically **multiplied** total AI compute usage across industries, as more organizations layered advanced AI-enabled services on top of legacy systems. Rather than reducing power consumption overall, the sudden expansion of cheap, powerful reasoning tools led to *greater* aggregate electricity demand. In effect, DeepSeek R2 illustrates how efficiency gains, coupled with open-source accessibility, can **accelerate** AI's growth curve—ultimately driving higher energy requirements despite per-task improvements.

Experiential AI and Robotics – New Sources of Demand: Beyond the cloud, the next frontier is AI systems operating in real time in the physical world. Consider autonomous vehicles (AVs). A self-driving car must constantly process sensor

data (cameras, lidar, radar) and run AI algorithms to navigate safely – effectively **a data center on wheels**. This requires substantial onboard electricity. A recent review finds that current autonomous driving systems draw on the order of **500 watts up to 2,500 watts (2.5 kW) per vehicle** for computing and sensors, depending on the level of automation and hardware used. Waymo’s self-driving system, for example, consumes about **1 kW of continuous power** just for the AI “brain”. For an electric vehicle, dedicating 1–2 kW to the AI stack can noticeably reduce driving range (since that power is diverted from propulsion). Now imagine millions of autonomous EVs on the roads in the coming decades – this implies **GW-level loads** collectively just to run vehicle intelligence.

Similarly, **robotics and automation** in warehouses, factories, and even homes will add to energy demand. Today’s warehouses use fleets of mobile robots for inventory and fulfillment; each robot requires battery charging and power for motors and onboard processors. As companies deploy more robots (and more advanced AI-driven robots that can operate independently 24/7), their electricity use grows. One can view these robots as new electric “workers” – analogous to adding more machines during the Industrial Revolution – which raises the facility’s power requirements. Humanoid robots under development (e.g. Tesla’s *Optimus* robot) are envisioned to perform general tasks in human environments. If such devices become widespread, **every additional 100,000 robots could represent several MW of added demand** (assuming each might consume on the order of hundreds of watts to a kW when active, plus charging losses). While still nascent, **the trend toward AI in embodied forms means energy consumption is no longer confined to climate-controlled data center racks, but also distributed across vehicles, warehouses, and cities**.

Edge AI and IoT: Another aspect is the proliferation of intelligent IoT (Internet of Things) devices – from smart sensors to AR/VR systems – that use AI at the **edge** (on-device processing). Edge AI can reduce data center load by handling computations locally, but it shifts some energy use onto devices and network equipment. For example, a smart security camera with onboard AI uses more power than a dumb camera, because it’s constantly running vision algorithms. Multiply this by billions of smart devices (in homes, retail, healthcare, etc.), and it becomes a non-negligible part of the energy puzzle. Moreover, the **data generated by experiential AI (vehicles, drones, etc.) is enormous**, and uploading this to the cloud for further analysis or machine learning will tax network infrastructure and data center storage, again indirectly consuming more energy.

In summary, as AI evolves “**from data to experience**,” its energy needs diversify. **Cloud AI (training and big-data analytics)** will keep drawing exponentially growing power in centralized facilities. **Distributed AI (autonomous machines and devices)** introduces entirely new loads on transportation and industrial energy budgets. The common denominator is that **electricity demand is set to rise on multiple fronts** – in core networks, in edge devices, and in the electrification of systems that were not electric before (e.g., replacing human labor or gasoline vehicles with electric, AI-driven alternatives). This multifaceted growth of AI’s energy footprint is already beginning to show up in global and national energy statistics, which we explore next.

Global Energy Demand Forecasts and Gaps

How exactly will AI’s growth translate into global and U.S. energy demand over the coming years? Analysts are grappling with this question, and while precise numbers vary, the consensus is that **AI will significantly amplify electricity demand beyond**

“business-as-usual” trends. In this section, we review key projections from major agencies and consultancies (IEA, EIA, McKinsey, BCG, etc.) and highlight the emerging gap between conventional forecasts and the potential AI-driven trajectory.

Baseline Global Projections: Prior to the recent AI boom, energy forecasts already anticipated notable growth, especially in electricity. The International Energy Agency’s mid-2020s outlook sees world electricity consumption rising at its fastest clip in recent memory – about **3.7% per year through 2027**. This equates to adding **over 900 TWh of new demand annually**, akin to injecting an economy the size of Japan’s power consumption into the grid each year. Drivers for this baseline growth include industrial production, increased air conditioning in developing countries, transport electrification (EV adoption), and the **expansion of data centers**. Notably, even without AI-specific considerations, **data centers were already flagged as a rapid growth segment** – alongside EVs and cooling – in the IEA’s analysis.

Long-term, McKinsey's *Global Energy Perspective 2024* report provides scenario-based estimates. In a middle-ground scenario ("Continued Momentum"), **global final energy demand grows ~11% by 2050**, whereas in a slower transition scenario it could grow ~18% by 2050. Much of this increase is concentrated in emerging economies as living standards and electrification rise. Importantly, these scenarios **assume certain efficiency gains and a moderate pace of new demand sources**. McKinsey notes that **new sources of demand and smaller-than-expected efficiency improvements could cause demand to swerve upward unexpectedly**. AI certainly qualifies as a *new source of demand* that wasn't fully accounted for in many pre-2023 forecasts.

Data Centers: Doubling or More by 2030: Over the past year, a number of analyses have zeroed in on data center electricity usage as a proxy for AI's impact. Key findings include:

- **IEA (2025)** – The IEA's April 2025 *Energy and AI* special report projects that **global data center electricity demand will more than double by 2030, reaching around 945 TWh** (from roughly ~450 TWh in 2022). Crucially, ****AI-specific workloads are expected to quadruple data centers' power draw in that period**. By 2030, powering data centers will require as much electricity as a medium-sized country (Japan-level consumption). The IEA highlights that **AI is the "most significant driver" of this surge**, potentially reversing a prior trend where efficiency kept data center energy flat despite rising workloads.
- **Goldman Sachs (2023)** – Goldman Sachs Research similarly estimates a **+160% increase in data center power demand by 2030** due to AI. Data centers, which today consume ~1–2% of global electricity, could reach **3–4% of global consumption by decade's end**. In the U.S. and Europe, this implies a return to electricity demand growth "*not seen in a generation*" as these regions' power use had plateaued. GS analysts note that from 2015–2019, data center workloads tripled but power use stayed flat ~200 TWh due to efficiency gains; however, **since 2020 those efficiency gains have slowed, and usage is rising sharply**. By 2028, they expect **AI-related processes to constitute ~19% of data center electricity demand**.
- **Boston Consulting Group (2024)** – BCG reports that **global data center electricity usage jumped 72% from 2019 to 2023**, largely attributable to AI adoption. Looking ahead, they warn that **by 2027 it could double again**, reaching ~2.6% of world electricity (from ~1.4% today). BCG emphasizes that beyond 2027, **projections become highly uncertain** – depending on the pace of generative AI uptake, regulatory constraints, and infrastructure expansion – but the trajectory points to a "new era in energy consumption, where AI doesn't just demand innovation – it drives it".
- **McKinsey (2023)** – Focusing on the U.S., McKinsey estimates that **U.S. data center electricity demand will rise by ~400 TWh from 2024 to 2030 (23% CAGR)**, representing **30–40% of all new electricity demand added in the country** in that period. This is striking given that overall U.S. power demand has been nearly flat since 2007. In effect, AI and cloud growth could single-handedly put U.S. electricity demand back on a strong growth curve, requiring tens of gigawatts of new capacity. McKinsey notes achieving just a quarter of AI's economic potential by 2030 would require **50–60 GW of new data center capacity in the U.S. alone**. Similar trends are noted in **Europe**, where data centers (propelled by AI and digitization) are expected to almost triple their power use to >150 TWh by 2030 (from ~62 TWh now) – reaching ~5% of Europe's electricity consumption (up from ~2% today).

Beyond Data Centers – Total Demand Effects: It's important to stress that these data center projections are a *component* of overall demand. In addition, advancements like DeepSeek actually are forcing increased predictions by many on needs due to increasing usage for higher compute needs faster than previously expected. Other AI-related electrification will add on top. For instance, greater **EV adoption (some attributable to autonomous fleets)** increases electricity use for transport. If autonomous vehicles make mobility more accessible or efficient, vehicle miles traveled (VMT) could increase, boosting electricity demand for EV charging (even as gasoline demand falls). Smart buildings with AI optimization might save some energy, but widespread use of electric HVAC and AI-driven climate control could paradoxically raise load if buildings aim for more data-center-like reliability and cooling for equipment. All said, **the net effect of AI across sectors is expected to raise the trajectory of energy demand relative to a non-AI scenario**.

The U.S. Department of Energy has explicitly recognized this shift. In mid-2024, DOE noted that after years of flat demand, **the U.S. is "returning to a period of rising electricity demand" due to transformations like AI, data centers, electrified transport and manufacturing**, projecting **total U.S. electricity demand growth of ~15–20% in the next decade**. It also reaffirmed the longer-term goal of **doubling electricity demand by 2050** as part of economy-wide

decarbonization – which implies massive investments in generation. If AI accelerates demand faster than expected, meeting that need while trying to hit climate targets becomes even more challenging.

Potential Gaps and “High Case” Scenarios: A critical question is whether current infrastructure plans and policy scenarios account for this AI-driven surge. There is a growing sense of a **gap between forecast and reality**:

- The IEA cautions that there are *significant uncertainties*, but their analysis puts advanced economies’ power sectors “back on growth footing” after years of stagnation. In the U.S., as noted, data centers could make up nearly **half of demand growth to 2030**, something not fully baked into many utility resource plans made just a few years ago.
- Europe’s grid operators, likewise, face a new reality. Europe’s overall power demand had been flat or even falling (due to efficiency and outsourcing of industry) from 2010s into early 2020s. The McKinsey Europe report underscores that **meeting data center (and AI) demand will require an “extensive increase” in electricity supply and grid upgrades – a notable shift for a region that hasn’t seen demand growth in 15+ years**.
- Some analysts have posited “high demand cases” where AI adoption is faster or deeper.

For example, if generative AI becomes ubiquitous in personal and business workflows (analogous to a new computing platform) or if autonomous electric vehicle rollout is very rapid, **2030 demand could overshoot even the upper ranges of current forecasts**. A Columbia University Energy Policy analysis (2023) estimated the additional load from generative AI under various usage scenarios, finding that in high-adoption cases the incremental power need could be comparable to that of a new major end-use sector emerging (hundreds of TWh globally).

In summary, **global and U.S. electricity demand is set to grow robustly through 2030, and AI is emerging as a key factor intensifying this growth**. Baseline forecasts (3–4% global growth, ~1–2% U.S. growth annually) may prove too conservative if AI computing needs accelerate. There is potential for an era of *re-synchronized* energy demand growth in advanced economies – something not seen since the early 2000s – largely attributable to digitalization and AI. This creates both an opportunity (for investment in new capacity) and a risk (if supply doesn’t keep up, leading to shortages).

The next section examines the supply side: how current energy policies and underinvestment trends might constrain the ability of the energy sector to meet this booming demand.

Energy Policy and Infrastructure Constraints

Even as AI turbocharges energy demand, the world’s ability to supply that energy reliably is under pressure from another historic shift: the drive toward cleaner, more sustainable energy. Global and U.S. energy policies in recent years have emphasized decarbonization – incentivizing renewables, discouraging fossil fuels – and this has led to **underinvestment in traditional energy infrastructure** relative to potential future needs. Here we evaluate how these policy constraints could create **supply-demand imbalances** over the next decade, especially in a scenario where demand outruns expectations.

Climate Policies and Fossil Fuel Investment Cuts: In 2021, the IEA made headlines by asserting that, under a Net Zero Emissions by 2050 scenario, *no new oil and gas fields* should be developed beyond those already approved, to limit warming to 1.5°C. While this was a normative scenario, some interpreted it as a call to drastically curtail fossil investment. Many governments and investors have indeed shifted capital away from fossil fuel projects, driven by climate commitments and ESG pressures. Upstream oil and gas spending fell sharply after 2014 and again in 2020, and has only partially recovered. **Global upstream investment was about \$499 billion in 2022**, still below the 2010s average in real terms. A joint IEF (International

Energy Forum) and S&P Global report estimates that **to meet projected demand through**

2030, upstream oil & gas investment needs to reach ~\$640 billion per year by 2030 – a 28% increase over current levels. That implies a **cumulative \$4.9 trillion by 2030** in oil/gas upstream capex is required to avoid supply shortfalls. Right now, we are not on track to hit that level of investment.

The risk of underinvestment is echoed by industry leaders. **Saudi Aramco's CEO Amin Nasser** has repeatedly warned that “**chronic underinvestment in hydrocarbons will keep global supply tight**”, and that assuming a rapid phase-out of oil/gas is dangerous. At a 2024 conference, he noted global oil demand hit a record 104 million barrels/day and argued “we should abandon the fantasy of phasing out oil and gas” prematurely, urging policymakers to **ensure sufficient investment to meet realistic consumption**. OPEC's outlook similarly projects that oil demand will **grow into the 2040s**, requiring continued upstream development; OPEC officials call underinvestment “a dangerous threat” to energy security if the world swings too far toward optimistic transition scenarios.

Electricity Grid and Generation Constraints: It's not just oil and gas. Power generation capacity investments are also running up against policy and permitting issues. Many countries are retiring coal plants (for climate reasons) and some nuclear plants (for safety/age reasons) while hoping renewables will fill the gap. However, renewables like wind and solar, while growing fast, face limits in terms of intermittency and land use. In the U.S. quite a few planned gas-fired plants have been canceled or delayed due to opposition or uncertainty over future climate regulations. **Transmission infrastructure** is another bottleneck – getting approvals to build high-voltage lines can take 5-10 years, slowing the integration of new renewable projects.

The **current grid** in both the U.S. and Europe was often cited as already struggling to meet reliability standards while integrating a rising share of renewables. BCG notes that the **trade-offs among reliability, cost, and sustainability are becoming acute**: trying to decarbonize (lower carbon energy) without compromising reliability is difficult if demand surges unexpectedly. California and Texas have seen instances of tight power supplies and even blackouts, partly because of extreme weather but also because capacity margins are thinner as old plants retire.

Now add AI-driven demand on top: **if data centers or new electric loads grow faster than planned, the grid could be caught short**. Already, local effects are visible – regions with major data center growth (Northern Virginia in the U.S., Dublin in Ireland, etc.) have faced grid constraints. In Northern Virginia (the world's largest data center hub), the local utility had to warn of potential capacity shortfalls by mid-2020s due to the sheer concentration of new data center projects drawing power. In Ireland, EirGrid (grid operator) at one point paused new data center grid connections to ensure the system could handle them, given that data centers could account for **up to 25% of Ireland's electricity demand by 2030** on current trajectories.

The Next Decade: Potential Imbalances: Over the next 5–10 years, several factors could stretch the supply-demand balance:

- **Electricity Supply Mismatch:** If AI and electrification push power demand above forecasts, we may see **capacity shortfalls** in regions that have been retiring fossil plants. The DOE has already highlighted the need to “**accelerate new investments in electricity generation and grids**” to accommodate AI's rise. If policy-driven plant closures outpace new additions, reserve margins fall. This leads to price volatility and possibly the need to restart or keep online some fossil units (as occurred in Europe when gas supply was tight in 2022 and coal plants were temporarily revived).
- **Fuel Supply Crunch:** On the fuel side, **natural gas** is a critical bridge fuel for electricity and heating. Underinvestment in gas production or pipelines (due to climate concerns or local opposition) could lead to **price spikes in gas**, especially if electricity producers suddenly need more gas to feed new data center loads or charging infrastructure. Europe learned this in 2021-2022 when gas supply constraints (exacerbated by geopolitical issues) caused record power prices. If new AI-driven demand appears in a context of constrained gas supply, it could similarly drive up costs.
- **Oil Demand and Refining:** AI itself doesn't directly burn oil (since it uses electricity), but indirectly a booming digital economy could sustain higher oil demand via freight, materials, or economic growth effects. Some oil executives warn that by 2028–2030, the world could struggle to produce enough oil if current low investment continues – leading to \$100+ per barrel prices or even physical shortages. Refining capacity is also being rationalized in some regions; a mismatch could cause fuel supply crunches (e.g., not enough diesel for backup generators that many data centers rely on for emergency power, ironically).
- **Mining and Materials:** Another angle is the materials needed for the energy transition: batteries (lithium, cobalt), power cables (copper, aluminum), wind turbines (rare earths, steel). Underinvestment in these supply chains, due to long lead times and environmental hurdles, could slow the deployment of renewables and storage that we *need* to meet

AI-driven demand sustainably. For example, a shortage of transformers or high-voltage equipment (a current concern in the U.S.) can delay grid expansion.

In conclusion, there is a real possibility that **policy and investment trends – aimed at sustainability – inadvertently create a short-term capacity crunch** just as AI-driven demand takes off. The 2020s could see periods of tight energy markets, with **higher and more volatile prices**, if supply lags behind. This dynamic would have wide economic implications:

energy-intensive sectors could suffer, inflation could be reignited, and governments might be forced to choose between climate targets and keeping the lights on. Ensuring that policy balances long-term decarbonization with near-term reliability is therefore critical. The silver lining is that such pressures could spur faster innovation in energy tech (e.g. energy efficiency, advanced nuclear, grid storage) as the system races to keep up – essentially, **AI might “stress test” the energy transition**, exposing weaknesses that need addressing.

Next, we identify how various segments of the energy industry – from fossil fuels to renewables – might fare in this environment, and which are poised to benefit or shift strategy due to AI’s influence.

Sector-by-Sector Impact on the Energy Complex

The coming decade’s shifts will reverberate across the entire energy value chain. In this section, we break down the **impacts on key energy sectors and technologies** – including natural gas, oil, coal, nuclear, renewables, and energy storage – highlighting which are likely to benefit from AI-driven demand growth and which may face challenges. We also discuss how a prolonged period of tight supply (if it materializes) could reshape investment strategies within the energy industry itself.

Natural Gas: The Critical Balancing Fuel – Natural gas stands out as a near-term beneficiary of rising electricity needs. Gas-fired power plants offer dispatchable generation that can rapidly scale to meet new loads (like sudden data center clusters) and back up intermittent renewables. The IEA projects that to meet surging data center demand, **gas and renewables will take the lead in key markets due to cost and availability**. We’re already seeing this: many large data centers contract directly for renewable power, but during peak usage or lulls in wind/solar, it’s gas plants that fill the gap. **Gas demand for power generation could thus see an uptick**, reversing what in some regions was a plateau or decline. Additionally, if AI demand stresses grids, some industries may install on-site gas generators for reliability, further supporting gas consumption.

For gas producers and infrastructure companies, this is an opportunity – higher utilization of gas pipelines, strong LNG demand (as countries without enough domestic gas import more), and potentially improved public perception of gas as a “keeper of grid reliability.” However, as noted, underinvestment is a risk. Regions like Europe have limited domestic gas and are trying to reduce dependence; if AI demand forces continued high gas use, Europe might need to lock in more LNG contracts (benefiting U.S. and Qatari LNG exporters). U.S. natural gas, with its cost advantage, could see robust market conditions if new power plants or LNG export terminals ramp up to serve AI’s indirect needs.

Oil and Refined Fuels: Indirect and Mixed Effects – Unlike gas, oil is not heavily used for electricity (only about 3% of global power). So AI’s direct effect on oil demand is limited. That said, robust economic growth fueled by AI productivity gains could bolster overall oil demand (for transport of goods, petrochemicals, etc.). Additionally, if autonomous vehicles increase total miles traveled (because of robotaxis, etc.), that could either increase electricity demand (if EVs) or oil demand (if they are hybrids or if autonomy also touches long-haul trucking which still uses diesel in near term). On the flip side, greater electrification and efficiency might erode oil usage in some sectors (e.g., AI-optimized logistics could reduce wasted trips, autonomous EVs displace gasoline cars).

Net-net, most forecasts (including OPEC’s) still see **oil demand growing through the 2030s** albeit at a slowing pace. AI likely won’t change the near-term oil demand trajectory drastically; it’s more about whether we hit peak oil demand in the 2030s or not. From an investment standpoint, if underinvestment continues and demand surprises to the upside, oil prices could stay structurally higher, benefiting upstream producers and oilfield services in the short-to-medium term. But beyond that, the sector remains challenged by the energy transition.

Many oil companies are hedging bets by investing in natural gas, hydrogen, or renewables – or by focusing on petrochemicals where demand is expected to persist. **AI could indirectly boost petrochemical demand** (for instance, data center components, plastics for devices, etc.), offering some support to the oil-refining complex oriented toward those products.

One area oil might see direct benefit is **backup power systems**. Data centers and critical AI infrastructure require highly reliable power, often maintained by diesel generators and fuel storage on-site. As more data centers are built, **diesel genset sales and fuel use for backup power** could increase (unless cleaner battery backups replace them). Companies providing backup power solutions (gensets, fuel supply contracts) might thus see growth as part of the AI infrastructure boom.

Coal: Diminished Role, But Not Zero – Coal is generally on the losing end of both AI trends and policy trends. Most new power demand will be met by gas and renewables, not coal, in OECD countries. Developing countries in Asia and Africa are the wildcard: they have growing demand and sometimes opt for coal due to local availability. If AI data centers expand in a country with a coal-heavy grid (say India or South Africa), in the short run that *could* increase coal burn. But globally, coal's share in electricity is expected to continue declining as older plants retire and very few new ones get built (China is still building some, but even they are investing heavily in renewables).

So, we don't expect AI to spark a coal renaissance; rather, coal could serve as a **stopgap in case of extreme shortages**. For example, if Europe or the U.S. faces blackouts, there might be political pressure to keep a coal plant on standby or delay its closure. Investors have largely moved away from coal miners and coal generation due to poor long-term outlook and ESG pressures. A possible short-term outcome of deficits is coal prices spiking (as seen in 2021) when gas is scarce, which can benefit the few remaining players, but this would likely be temporary. **In essence, coal might see transient upticks in use if things get tight, but it's not a growth sector** and will likely continue to shrink as a portion of the energy mix, even with AI-driven demand growth.

Nuclear: A Resurgence Opportunity – Nuclear power is uniquely positioned as a zero-carbon, high-reliability source of baseload electricity – exactly what a digital, AI-powered economy needs in the long run. After decades of stagnation in many countries, there are signs of renewed interest in nuclear due to climate concerns and energy security. AI's demand surge adds another argument: **without nuclear, it may be very difficult to both meet demand and decarbonize**. Countries like France, the UK, China, and even the U.S. are investing in next-generation reactors (including small modular reactors, SMRs). If policies adjust to encourage more nuclear development (streamlining regulation, providing incentives), the 2030s could see a wave of nuclear projects.

In the near term, existing nuclear plants are extremely valuable assets. Extending the life of current reactors (as the U.S. is exploring for several plants) would help supply stability. Nuclear companies (plant operators, reactor vendors like Westinghouse, EDF, Rosatom, etc.) could benefit from a more favorable policy environment and increased funding. From an investment theme perspective, **nuclear technology and related supply chain (uranium, reactor components) might see tailwinds**. For example, uranium prices have firmed up as more reactors are planned or life-extended. If AI's power hunger forces policymakers to rethink baseload capacity, nuclear could be one answer.

Renewables (Solar/Wind): Accelerated Deployment, Grid Challenges – Renewable energy will undoubtedly be a big winner quantitatively, as virtually every scenario relies on massive additions of solar and wind to meet new demand and climate goals. Already, renewables are projected to approach half of global power generation by 2030. AI's demand surge could further boost investments: more corporate PPAs (power purchase agreements) as tech firms and data centers seek power, more government support to ensure enough energy is available.

For instance, **Europe's data center growth will "require a big rise in electricity supply mostly from low-carbon sources"**, meaning wind farms in the North Sea, solar farms in

Spain, etc., all see increased demand for their output. The **Inflation Reduction Act (IRA) in the U.S.** already unleashed a boom in renewable project development; the need for AI-related power only reinforces that trajectory. Renewable developers and manufacturers (solar panel makers, wind turbine producers) are likely to experience high demand. Energy investors might find opportunities in these areas, as well as in ancillary services (engineering, construction, grid integration software).

However, integrating a high share of renewables comes with issues: intermittency and **the need for storage and grid enhancement**. As more of the load becomes critical (e.g., large data centers can't have outages), simply having a lot of renewables is not enough – you need them to be backed by storage or dispatchable sources. This brings us to...

Energy Storage and Grid Management: Rising Importance – **Utility-scale battery storage**, pumped hydro storage, and other storage tech (like new chemistries or gravity storage) will be crucial to smooth out the variability of wind and solar. The more we lean on renewables to power AI 24/7, the more storage is needed to guarantee reliability. The **energy storage sector** is therefore a clear beneficiary: global battery storage installations are projected to grow exponentially. Companies making batteries (LG, Tesla, Panasonic, etc.), those focusing on grid-scale solutions (Fluence, Wärtsilä), and even emerging longer-duration storage startups could see a significant market.

In addition, **grid infrastructure and smart grid tech** get a boost. More demand means more transmission lines to connect new generation sources, more substations, and more grid-hardening investments (to prevent outages). Also, **demand-side management and efficiency technologies** could gain traction – AI might actually assist here, as AI can optimize energy usage patterns (for example, shifting flexible loads to off-peak times). So there's a bit of a virtuous cycle: AI causes an energy problem, but AI can also be part of the solution by improving how we manage energy. Companies that offer grid analytics, AI-driven efficiency for buildings and factories, or virtual power plant software (aggregating backup generators or EV batteries to support the grid) may find growing markets.

Prolonged Deficits and Strategy Shifts: If supply deficits do occur and persist (say power shortages or high prices over several years), the energy industry's strategies will adapt. We could see:

- **Reinvestment in Capacity:** Energy companies (including oil & gas majors) might increase capital expenditure in core business – e.g., more drilling or faster gas project development – given higher price signals. Already some oil majors have scaled back renewables ambitions to refocus on profitable oil/gas projects as prices rose in 2022. If they sense that the world actually needs more hydrocarbons for longer, they may double down on core competencies. On the power side, utilities might propose new “all of the above” generation investments (including nuclear SMRs, gas with carbon capture, etc.) if regulators allow a broader mix to ensure reliability.
- **Energy Security Alliances:** Countries may forge deals to secure energy: for example, long-term LNG contracts (as Germany did after 2022), or partnerships to build renewable energy in one country to supply another (like North Sea offshore wind consortia). This could accelerate cross-border energy projects. It could also shift trade flows – e.g., U.S. LNG and Middle Eastern oil might have even larger market share if others cut back production too quickly.
- **Innovation Boost:** High energy prices and tight supply conditions historically spur innovation. We could expect a heightened push for breakthrough technologies: advanced battery chemistries (to store days of power, not just hours), *green hydrogen* (to store renewable energy seasonally or fuel industry), carbon capture (to enable more fossil use without the emissions penalty), and maybe fusion energy in the longer term. Governments might increase R&D funding for these in response to an energy crunch blamed on underinvestment.
- **Behavior and Policy Adjustments:** In a scenario of persistent deficits, policy might pivot to ensure supply – e.g., easing permitting for energy projects, providing capacity payments to keep certain power plants online, or even strategic reserves for gas as we have for oil. The political narrative could shift from purely “keep it in the ground” to “invest in all forms of energy responsibly.”

For investors, the implication is to be agile and watch policy signals. Energy companies that demonstrate flexibility – those that can pivot between energy sources, or integrate across the value chain (like oil majors getting into power utility business, or renewables developers adding storage to projects) – will likely outperform in a volatile environment.

In summary, **the energy complex is set for a period of re-pricing and re-balancing**. Sectors enabling more electricity (gas, renewables, nuclear, storage) stand to gain the most from AI's rise. Hydrocarbons face a complex picture: immediate upside if demand outstrips supply, but long-term questions about fuel applications remain. The key for all players will be to ensure resilience – having the right mix of assets to supply an electrifying world reliably and sustainably. Those that invest wisely in capacity and innovation may capture outsized returns, while those that stick blindly to outdated paradigms (either clinging only to fossil or only to intermittent renewables without backup) could struggle.

Next, we translate these sectoral trends into concrete **investment risks and opportunities**, and discuss strategic moves for investors looking at the intersection of AI and energy.

Investment Risks and Opportunities

For business leaders and investors, the rapid intertwining of AI growth and energy demand presents a new landscape of risks and opportunities. The value chain spans from the tech sector (which needs reliable power for AI) to the energy sector (which must deliver that power) and everything in between (infrastructure, equipment, materials). In this section, we outline key investment implications, identify who stands to win or lose, and suggest themes to focus on.

Opportunities: Who Stands to Benefit

- **Electric Utilities and Power Generators:** Companies that generate and supply electricity could see revenue growth as volumes sold increase. *Utilities* in regions with growing demand (e.g., those serving data center hubs or high-tech corridors) may justify new investments in rate base (new plants, grid upgrades), potentially boosting earnings. *Independent power producers (IPPs)* who can quickly add capacity (like new gas peakers or renewable farms) will find ready buyers for their electricity. However, the regulatory environment is key – those in markets that allow capacity payments or long-term contracts for new power will benefit most. Utilities that proactively invest in reliability (to avoid outages for key customers) can also gain competitive advantage in attracting data center projects to their area.
- **Natural Gas Producers & LNG:** As discussed, gas demand is highly likely to get a second wind. *Upstream gas producers* (especially low-cost shale gas players in the U.S. or LNG exporters) could profit from sustained or rising gas prices and volumes. *LNG infrastructure developers* (Cheniere Energy, for example) already see strong demand for export capacity; more global interest in gas for power means their facilities stay fully booked, and new expansion trains become viable. Additionally, companies involved in gas distribution and pipelines might see increased utilization; midstream MLPs (master limited partnerships) in the U.S. could have steadier, even growing, throughput. Gas producers like Diamondback and Chevron are also announcing behind-the-meter power arrangements, reflecting a shift toward more localized, flexible energy delivery models that complement traditional grid and export infrastructure.
- **Renewable Energy and Equipment Makers:** The push for clean power will benefit the entire renewables ecosystem. *Solar and wind farm developers* will have bigger markets (corporate PPAs with tech firms, utility procurements, etc.). Manufacturers of *solar panels, wind turbines, and related components* should see volume increases. For example, if data center companies commit to 100% renewable energy, they often finance new solar/wind installations. The IRA in the U.S. also encourages domestic manufacturing of renewable tech – good news for solar supply chain investments. Plus, *inverters, transformers, and grid gear* needed to integrate renewables are in high demand (e.g., companies like SolarEdge or Siemens Energy that make such equipment could benefit).
- **Energy Storage & Battery Value Chain:** As noted, *battery manufacturers and storage project developers* stand to gain as storage becomes essential. The lithium-ion battery supply chain (from mining companies like Albemarle (lithium) to battery cell producers and integrators) has strong secular growth. Beyond batteries, any technology that can store energy – compressed air storage, flow batteries, etc. – if proven at scale, could fill a huge need. There might be opportunities in *mining and materials* as well: lithium, cobalt, nickel for batteries; copper for all the electrification (wiring, motors); rare earth elements for wind turbine magnets and EV motors. Investors could look at commodity plays or specialized mining companies positioned to supply the raw materials of the electrified economy.
- **Grid Infrastructure and Services:** Companies specializing in *transmission lines, smart grid software, and grid construction* should see a boom. Governments are allocating funds for grid modernization (e.g., the U.S. Infrastructure Act). Firms that build high-voltage lines, or provide HVDC (high-voltage DC) solutions to connect renewable resources to load centers, are crucial. Also, *engineering and construction (EPC) firms* with energy project expertise will have full order books (building new plants, terminals, etc.). Even utility-focused tech companies – those making grid automation systems, advanced meters, and AI for grid management – have a growing market as utilities digitalize their networks to handle complexity.
- **Data Center and Cloud Providers (Energy-Savvy Tech):** Interestingly, the tech companies driving AI are themselves adapting. The **hyperscale cloud providers (Amazon, Microsoft, Google)** are investing in their own energy

infrastructure – signing huge renewable deals, developing on-site generation and backup. They might not directly be profit opportunities in energy (since energy is a cost center for them), but those that manage to secure cheap, power will have an edge in offering AI services at lower cost. Tech companies that become leaders in energy efficiency (e.g., designing chips that consume less power per computation, like Google's TPUs or Nvidia's efficiency improvements) will also stand out. **Semiconductor companies** focusing on AI accelerators have incentive to prioritize performance per watt; if they succeed, they capture AI market share (Nvidia is a prime example whose valuation soared as their chips enable AI, though they also drive energy use).

- **Energy Efficiency and AI Solutions:** As energy becomes a bigger cost, *companies offering energy-efficient solutions* should gain traction. This includes makers of advanced cooling systems for data centers (e.g., liquid cooling to reduce power for cooling), companies doing AI-driven optimization of industrial processes (to cut energy waste), and even firms retrofitting buildings for efficiency (insulation, HVAC upgrades) as power costs rise. AI itself can be employed to make other sectors more efficient – a growth area for AI startups working on energy optimization (like AI for better grid balancing, or for reducing idle server power draw, etc.). Investors may find opportunities in such cleantech/AI crossovers.
- **Oil & Gas (Short/Mid Term):** Traditional oil and gas companies could enjoy strong cash flows in the short to medium term if energy markets stay tight. We've seen in 2022–2023 record profits for oil majors when prices spiked. If underinvestment leads to a structural supply gap, these companies could see prolonged periods of high commodity prices. Many are returning cash to shareholders through dividends and buybacks, which could make them attractive from a pure return perspective. They are also cheaply valued and underowned. The key is timing. In the meantime, some oil companies with gas-heavy portfolios or integrated operations (including trading) might be relatively well positioned. Also, those that invest in **carbon capture and storage (CCS)** to keep their operations viable in a carbon-constrained world may open new revenue streams (e.g., selling carbon credits or using CO₂ for enhanced oil recovery).
- **Nuclear and Alternative Baseload:** *Nuclear energy companies* – whether utilities operating plants or startups developing SMRs – could see growing investor interest. Countries like the UK and France offering capacity contracts or incentives for new nuclear make those projects potentially lucrative. There are also publicly traded companies in the nuclear fuel cycle (uranium miners like Cameco, nuclear tech firms like BWX Technologies that build components). Outside of conventional nuclear, if *geothermal energy* can be tapped (e.g., using AI to locate better drilling sites or new tech for deep geothermal), it provides another baseload-like renewable source. Companies exploring geothermal or other firm renewables could benefit from a push for 24/7 clean power.

Risks: What to Watch Out For

- **Energy-Intensive Industries:** Sectors that consume a lot of energy as input face rising cost pressures. Think of **chemicals, steel, aluminum, cement, mining, manufacturing** in general – if electricity and fuel prices go up due to tight supply, their operating costs increase. Unless they can pass costs through (which depends on market power), margins shrink. For example, data center operators themselves – if they didn't lock in long-term power contracts – could see cost spikes (though many do hedge via PPAs). Similarly, *microchip fabrication* is extremely electricity-intensive; higher power costs could make chip production pricier or push it to regions with cheaper energy. Investors in such industries need to evaluate energy hedging strategies and energy efficiency measures of those companies. Some companies might relocate operations to regions with more abundant energy (like moving a factory from a high-cost electricity country to a lower-cost one, which can have geopolitical and societal implications).
- **Utilities and Obsolescence:** While utilities benefit from demand growth, they also face **technological risk**. If they invest heavily in new gas fired or coal fired capacity and then later climate policies or new technologies force early retirement, those assets could become uneconomic. There is regulatory risk as well – if high costs to maintain reliability lead to significantly higher customer rates, public utility commissions might disallow some costs or push for alternative solutions. Utilities must also manage reliability or face fines and reputational damage (e.g., prolonged blackouts anger regulators and the public). So picking the right utilities – those in supportive regulatory environments and with balanced portfolios – is important.
- **Policy & Regulatory Risk:** The interplay of policy is complex. If energy prices remain higher for longer, governments might intervene (windfall profit taxes on energy companies, price caps like some European countries attempted, stricter

efficiency or rationing rules). For instance, if data centers are seen as causing grid stress, we could see local moratoria or requirements for them to invest in their own renewable power or backup (increasing their costs). On the flipside, to encourage AI growth, some might subsidize energy for tech (as some states do to attract data centers). Constant policy changes can create uncertainty for investors. There's also climate policy risk: a sharp turn towards stronger climate action (due to extreme weather or political change) could impose carbon prices or mandates that penalize fossil fuel-heavy businesses quickly.

- **Technological Disruption:** From the energy side, a major breakthrough (say, **fusion energy commercialization** sooner than expected, or ultra-cheap long-duration storage) could upend the expected supply/demand calculus. If fusion became viable by late 2030s, for example, it could render concerns about energy scarcity moot in the long term (though that seems distant as of now). On the AI side, if AI hardware becomes radically more efficient (orders of magnitude improvements) or if there's a saturation point in AI adoption, the expected demand boom might moderate. Investors need to monitor R&D progress: both in energy tech and in AI chip design. Companies stuck with yesterday's technology could lose out (e.g., an older power plant that can't ramp quickly in a dynamic grid, or a data center full of inefficient servers when competitors built ultra-efficient ones).
- **Physical Climate Risks:** It's worth noting that increased energy infrastructure is also exposed to physical climate risk (extreme weather, droughts affecting hydro or cooling water for plants, etc.). More demand stresses an already climate-vulnerable grid. If climate change leads to more wildfires, hurricanes, heatwaves – these can knock out power or fuel supply and cause economic losses. Energy and utility investors are increasingly evaluating how resilient their assets are to such events. Failure to do so can lead to catastrophic losses (e.g., PG&E's bankruptcy due to wildfire liabilities).
- **Macro-economic Factors:** A scenario of persistently high energy costs could contribute to broader inflation and potentially slower economic growth, which indirectly affects all investments. Conversely, if AI boosts productivity substantially, it could increase economic growth – but that might lead central banks to hike interest rates sooner if inflation appears, impacting financing for large energy projects (which are capital-intensive). High interest rates make it more expensive to build new infrastructure (debt costs higher). So macro conditions, debt availability, and currency risks (for global projects) all interplay. For instance, emerging markets needing to invest in energy might struggle if their currencies weaken or if financing costs jump, potentially leading to project delays (and thus even tighter supply).

Thematic Investment Ideas

Bringing it together, here are a few **investment themes** likely to outperform due to AI-induced energy shifts:

1. **“Electrify and Amplify” – Grid and Power Infrastructure:** Companies central to expanding and upgrading the grid and power supply. Examples: utility ETFs or specific stocks in growth markets; manufacturers of HVDC equipment, transformers (which have long backlogs now); engineering firms like Quanta Services (involved in transmission build-out); copper future and ETFs (as copper is the wiring metal of choice).
2. **“Digital meets Sustainable” – New Energy for AI:** Investments at the nexus of big tech and clean power. For instance, funds or companies that develop renewable energy expressly for sale to corporates (e.g., Brookfield Renewable), or yieldcos that own renewables with long-term contracts from creditworthy tech firms. Also, datacenter REITs that power facilities with new energy and have strong energy management practices – they can charge premium rents to clients like AI companies if they offer reliability and sustainability.
3. **“AI Hardware Ecosystem” – Efficiency and Supply Chain:** This includes chipmakers focusing on performance-per-watt (Nvidia, AMD, specialized AI chip startups), advanced cooling companies (cooling is a significant part of data center energy use), and even software companies enabling more efficient AI workloads (like AI model optimization platforms). As energy becomes a constraint, whoever can do the same AI work with less energy will win clients – so investing in the “picks and shovels” that make AI more efficient is promising.
4. **“Reliability and Resilience” – Backup Power and Storage:** Firms that provide solutions for always-on power. Apart from batteries, think of companies like Generac (diesel and gas backup generators, now also into home batteries) – data centers and businesses might install more backup gensets or microgrids. Also, companies enabling microgrids (Schneider Electric, for instance, provides microgrid controllers and services) because critical facilities might not trust the main grid fully. In an AI-driven economy, downtime is costly, so money will flow into resilience.

5. **Commodities and Resource Hedge:** Given the uncertainty, some investors may choose to hold a basket of energy commodities or related equities (oil, gas, uranium, lithium, copper) as a hedge that at least some of these will be in high demand. Energy sector ETFs or resource-focused funds could provide broad exposure.
6. **Emerging Markets Power & Tech:** A lot of the increased energy demand will come from emerging markets (China, India, Southeast Asia, Africa) as they grow and also adopt AI. Companies that can capitalize on building energy and digital infrastructure in those regions may thrive. This could be multinational engineering firms, or local champions in power generation. Also, emerging markets utilities or IPPs might have higher growth rates than their developed counterparts (though also more risk).
7. **Energy Future Enablers:** These are companies bridging old and new energy – like those doing carbon capture (enabling continued use of gas for power with lower emissions), hydrogen production (which could store surplus renewable energy and later be burned in turbines for peak power), or grid-scale demand response aggregators (like companies that aggregate data center backup generators into a virtual power plant to sell power back to the grid at peak times). These might be smaller or private now, but could become significant.

Strategic Considerations for Investors

Investors should adopt a **balanced, portfolio approach** to this dynamic landscape:

- **Diversification across the energy spectrum** can hedge against the uncertainties of policy and technology. One could hold both energy growth stocks and some traditional energy value stocks to cover both outcomes (rapid evolution vs. slower scenarios).
- **Timeline alignment:** Shorter-term trades might favor oil & gas (if expecting a supply crunch in the next 2-5 years), whereas long-term investments should lean into nuclear, renewables, grids, and tech efficiency (the structural direction of the economy).
- **Active engagement:** Particularly for large investors, engaging with companies on their energy strategies can be fruitful. For example, encouraging a data center REIT to invest in solar panels or more efficient cooling could improve its risk profile. Or pushing an industrial company to lock in long-term renewable contracts could shield it from price volatility.
- **Monitoring indicators:** Keep an eye on indicators like data center power usage growth,

AI adoption rates in industry, capacity investment trends, and policy announcements (e.g., if a government announces a crash program to build SMRs or a ban on certain high-energy mining like crypto, etc.). These will inform which sectors are heating up or facing headwinds.

In conclusion, the intersection of AI and energy is creating a new paradigm for investors. There will be **clear winners: companies that supply the tools to power and efficiently run the AI revolution**, and **clear losers: those caught unprepared or on the wrong side of cost structures** in an energy-hungry world. By understanding the trends and prudently allocating capital, investors and businesses can not only mitigate risks but also ride the wave of this historic shift, much like those who invested in railroads during the steam age or in semiconductors during the digital age. The stage is set for significant capital reallocation – and potentially, significant returns for those who navigate it wisely.

Conclusion

The rapid growth of artificial intelligence is ushering in a new chapter of the global energy story

– one where **bits and bytes drive demand for watts and volts**. Our analysis illustrates that AI's ascent will likely amplify energy consumption beyond prior forecasts, posing both challenges and opportunities for economies, markets, and investors.

In many ways, history is repeating itself: just as the Industrial Revolution and the Information Age were enabled by abundant energy (coal and electricity, respectively), the AI revolution is shaping up to be constrained or empowered by

our energy infrastructure. Electricity is the lifeblood of AI, and ensuring sufficient, reliable power will be paramount to realizing AI's economic potential. The historical lesson is clear: when transformative technologies emerge, societies must quickly scale energy supply to match – or risk bottlenecks and instability.

The coming decade will test the resilience and adaptability of our energy systems. On the demand side, we can expect global electricity usage to surge, with data centers and device electrification at the forefront. AI-heavy sectors (from cloud computing to autonomous transport) will become major energy consumers in their own right. On the supply side, the twin pressures of climate policy (limiting emissions) and the need for growth (more power) will necessitate creative solutions and significant investment. A central risk is that policy overshoots in suppressing traditional energy before alternatives can fully take over, leading to supply-demand mismatches. Conversely, there is also the risk that the world doesn't move fast enough, undermining climate goals as AI pushes emissions back up. Steering between these outcomes will require nuanced strategy and cooperation between tech firms, energy companies, and policymakers.

For business and investment professionals, the takeaway is to **integrate energy considerations into AI strategies (and vice versa)**. Companies implementing AI at scale must factor in energy availability, costs, and sustainability – the era of treating electricity as an afterthought is over. Meanwhile, energy companies should view AI not just as a load growth factor but also as a tool – employing AI for grid optimization, predictive maintenance, and discovery of resources can enhance efficiency and output. Investors have the chance to fund the bridging of this gap: financing new power projects, backing startups that alleviate energy constraints, and holding companies accountable for forward-looking energy risk management.

In practical terms, we anticipate several developments:

- **Stronger public-private initiatives** to expand power generation and grid capacity, often with a focus on new energy, in order to accommodate AI and electrification.

Governments may label data centers and AI infrastructure as critical, streamlining their energy access and supporting R&D in energy-efficient computing.

- **Market signals for reliability** will grow – capacity markets, grid services payments, and long-term power contracts will proliferate, assigning value to being available when needed. This benefits those who invest in firm power capacity (gas peakers, hydro, storage, etc.) and penalizes pure intermittent generation without backup.
- **Innovation accelerates on both fronts:** AI research will likely devote more effort to *energy efficiency*, given cost and corporate governance pressures, potentially yielding algorithms that achieve the same outcomes with less computation.

Simultaneously, energy tech innovation (from advanced batteries to maybe AI-assisted nuclear reactor design) will get a boost under the urgent need to expand supply sustainably.

- **The investment narrative shifts:** We'll hear more about the **"AI-Energy nexus"** in earnings calls and investor conferences. Tech companies might highlight how they are securing power and improving efficiency as part of their growth story. Energy companies will talk about selling not just commodities, but solutions tailored for data and tech clients (e.g., "firm power for your AI data hub").

In closing, the rapid electrification driven by AI's rise could be a catalyst that ultimately **propels the energy innovation forward** – albeit not without bumps along the road. It shines a spotlight on the importance of modernizing infrastructure and balancing portfolios of energy sources. Those stakeholders that recognize and plan for this symbiosis of AI and energy will be better positioned to thrive. Much like steam engines needed coal, and computers needed silicon chips, **AI needs electrons** – and ensuring a robust flow of those electrons may well define the winners of the next economic era.

The world stands on the cusp of this interlinked transformation. For analysts and investors evaluating the space, the imperative is clear: **stay informed, stay agile, and align strategies with the fundamental reality that in the age of AI, energy is kingmaker.** By doing so, we not only safeguard economic growth and corporate success, but also guide capital toward building an energy system capable of supporting the bright, intelligent future that AI promises.

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